

Exact and asymptotic boundary integrals for SPH kernels: divergence-theorem edge reductions, closed forms and curvature expansions

curvatureBoundaries project

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Abstract

We collect, in one logical chain, the mathematics behind the boundary integrals used to couple smoothed-particle-hydrodynamics (SPH) fluids to solid boundaries: the kernel integral $\lambda(\mathbf{x}) = \int_S W(|\mathbf{x} - \mathbf{x}'|) d\mathbf{x}'$ over a solid S , its gradient, its moments, and the nodal weights of fields extended into the solid. We prove (i) closed forms for planar and spherical boundaries with Wendland and cubic-spline kernels, (ii) a divergence-theorem reduction of area integrals over polygons to one-dimensional edge integrals, whose integrands are elementary for piecewise-polynomial kernels, (iii) a compact-potential recursion for higher moments and its use for FEM nodal weights, (iv) a closest-point curvature expansion and a slender-body (mean-value) series for boundaries that are curved or thin compared to the support radius, and (v) the wall Laplacian and wall-viscosity operators built on the same objects. Every statement is tagged with how it was checked (symbolically in Maple, against independent quadrature, or not at all), and an appendix lists the problems found while auditing the existing note collection.

Contents

1	Setting and conventions	2
1.1	Kernels	2
1.2	The boundary integrals	3
1.3	The idea in one page	3
1.4	Notation	4
1.5	Verification tags	4
2	Planar and spherical boundaries	4
2.1	Three dimensions	4
2.2	Two dimensions	5
2.3	Solid ball in 3D	6
2.4	Consistency of the two routes	6
3	Edge calculus in two dimensions	6
3.1	Geometry	6
3.2	Elementary primitives	7
3.3	Truncated monomials	7
3.4	Value identity	7
3.5	Degenerate placements	8
3.6	Gradient identity	8
3.7	Compact-potential recursion for moments	9
3.8	Half plane and the planar closed form	10

4	Fields on elements: FEM nodal weights	10
5	Closed boundaries (data on the boundary only)	11
5.1	Polygonal approximation of a curved boundary	11
5.2	Limits	11
6	Curvature expansion of the value integral	11
6.1	Statement	12
6.2	Evaluation	12
6.3	Accuracy	13
6.4	General curves	13
7	Slender and thin obstacles	13
8	Extension to three dimensions	14
9	Wall operators built on the edge calculus	15
9.1	The wall Laplacian	15
9.2	The viscosity coefficient	16
9.3	Free-slip wall term	16
9.4	No-slip flux term	17
10	Verification ledger	17
A	Audit	17
A.1	The existing derivation notes	17
A.2	The first draft of this paper	18
A.3	Independent numerical verification of the 2D claims	19

1 Setting and conventions

1.1 Kernels

Throughout, $n \in \{2, 3\}$ is the spatial dimension and h the *support radius*. A kernel is

$$W(r; h) = \frac{C_n}{h^n} \widehat{W}\left(\frac{r}{h}\right), \quad \widehat{W} \in C([0, 1]), \quad \widehat{W}(q) = 0 \quad (q \geq 1), \quad (1)$$

with C_n fixed by $\int_{\mathbb{R}^n} W = 1$, i.e. $2\pi C_2 \int_0^1 \widehat{W} q \, dq = 1$ and $4\pi C_3 \int_0^1 \widehat{W} q^2 \, dq = 1$. We work at $h = 1$ and restore h by scaling (Lemma 1.2). The kernels used are listed in Table 1; all are piecewise polynomial in q .

kernel	$\widehat{W}(q)$	C_2	C_3
cubic	$(1 - q)^3 - 4(\frac{1}{2} - q)_+^3$	$\frac{80}{7\pi}$	$\frac{16}{\pi}$
Wendland C2 (w2)	$(1 - q)^4(1 + 4q)$	$\frac{7}{\pi}$	$\frac{21}{2\pi}$
Wendland C4 (w4)	$(1 - q)^6(1 + 6q + \frac{35}{3}q^2)$	$\frac{9}{\pi}$	$\frac{495}{32\pi}$
Wendland C6 (w6)	$(1 - q)^8(1 + 8q + 25q^2 + 32q^3)$	$\frac{78}{7\pi}$	$\frac{1365}{64\pi}$

Table 1: Kernels at $h = 1$ (Wendland [1]). On $[0, \frac{1}{2}]$ the cubic equals $\frac{1}{2} - 3q^2 + 3q^3$, so $\widehat{W}(0) = \frac{1}{2}$.

Lemma 1.1 (Normalisation). *The constants in Table 1 satisfy $\int_{\mathbb{R}^n} W = 1$ for $n = 2, 3$.*

Proof. Direct integration of polynomials; checked symbolically (Maple, `20_paper_audit.mpl`). $\square$

1.2 The boundary integrals

Let $S \subset \mathbb{R}^n$ be the solid and $\mathbf{x}$ the evaluation point (a fluid particle). With $\mathbf{y} = \mathbf{x}' - \mathbf{x}$ and $r = |\mathbf{y}|$ we study

$$\lambda(\mathbf{x}) = \int_S W(|\mathbf{x} - \mathbf{x}'|) d\mathbf{x}', \quad (2)$$

$$m_\alpha(\mathbf{x}) = \int_S \mathbf{y}^\alpha W(r) d\mathbf{x}', \quad |\alpha| = k, \quad (3)$$

$$g_\alpha(\mathbf{x}) = \int_S \mathbf{y}^\alpha \nabla_{\mathbf{x}} W(|\mathbf{x} - \mathbf{x}'|) d\mathbf{x}'. \quad (4)$$

The value λ is the volume of solid seen by the particle; m_α and g_α are needed for fields extended into the solid (Section 4). The distance from $\mathbf{x}$ to ∂S is $d \geq 0$ for a fluid-side particle ($d < 0$ inside the solid, $\lambda(-d) = 1 - \lambda(d)$ for a flat wall).

Lemma 1.2 (Scaling). $\lambda(d; h) = \lambda(d/h; 1)$ and $m_\alpha(d; h) = h^{|\alpha|} m_\alpha(d/h; 1)$, where d stands for the whole geometry measured in the same unit as h . Consequently $\nabla_{\mathbf{x}} \lambda$ carries a factor h^{-1} and $\partial_{\mathbf{x}} m_\alpha$ a factor $h^{|\alpha|-1}$.

Proof. Substitute $\mathbf{x}' = \mathbf{x} + h\mathbf{y}_1$ in Eqs. (2) and (3): $d\mathbf{x}' = h^n d\mathbf{y}_1$ cancels the h^{-n} in Eq. (1), and $\mathbf{y}^\alpha = h^{|\alpha|} \mathbf{y}_1^\alpha$. $\square$

1.3 The idea in one page

Everything below is one trick used three times: *write the integrand as a divergence, so the area (volume) integral over the solid becomes a boundary integral, and only the part of the boundary inside the support radius contributes.*

quantity	vector field whose divergence (or gradient) is the integrand	result
value $\int_T g$	$\mathbf{G} = \mathbf{y} M_g(r)/r^2$ with $\nabla \cdot \mathbf{G} = g$	Theorem 3.3
gradient $\nabla_{\mathbf{x}} \int_T g$	translate the domain: $\partial_\delta \int_{T-\delta \mathbf{e}} g = -\oint n g$	Theorem 3.4
moments $\int_T \mathbf{y}^\alpha f$	$\mathbf{y}^\beta \Phi[f]$ with $\nabla \Phi = \mathbf{y} f$	Theorem 3.7

For a polygon the boundary is a union of edges; on each edge the integrand is an explicit function of the single variable s (position along the edge), and for piecewise-polynomial kernels these one-dimensional integrals are elementary (Proposition 3.1). Sections 2–3 give the exact oracle and the edge calculus; Sections 4 and 5 use them for fields on elements and for closed boundaries; Sections 6 and 7 treat smooth or thin bodies asymptotically instead of adding edges; Sections 8 and 9 extend and use the results.

1.4 Notation

symbol	meaning
$\mathbf{x}, \mathbf{x}', \mathbf{y}, r$	evaluation point, integration point, $\mathbf{y} = \mathbf{x}' - \mathbf{x}$, $r = \mathbf{y} $
$h, q = r/h$	support radius of the kernel, scaled distance ($h = 1$ in all derivations)
d, z_e	distance of $\mathbf{x}$ to a wall (Section 2); signed distance of $\mathbf{x}$ to the line of edge e (Section 3), $z_e = \mathbf{n}_e \cdot (p_e - \mathbf{x})$
r_c	<i>cut radius</i> of a radial profile g ($g = 0$ for $r > r_c$); $r_c = 1$ for a full kernel, $r_c = \frac{1}{2}$ for the inner cubic block
$R, \kappa = 1/R$	radius and curvature of a disk or ball (Sections 2.3, 5 and 6); never a cut radius
$M_g(r)$	$\int_0^r t g(t) dt$ (2D) or $\int_0^r t^2 g(t) dt$ (3D), the <i>radial mass</i> of g
$\Phi[f], \varphi(\sigma)$	compact potential of a radial profile f (Theorem 3.7); the kernel as a function of $\sigma = r^2$ (Section 6)
$I_m(s)$	$\int_0^s r^m ds$ along an edge line (Proposition 3.1); not to be confused with the planar family $\mathcal{J}_m(d)$ of Lemma 2.4
λ_Δ	$\int_S \nabla_{\mathbf{x}}^2 W$, the wall Laplacian (Section 9); a number, not a difference
$\mathbb{I}[\cdot]$	indicator

1.5 Verification tags

Each result carries a tag: [M] checked symbolically in Maple (exact arithmetic, or 40–50 digit quadrature where stated); [N] checked numerically against an independent quadrature (tests/); [A] argued here but not independently checked; [P] plan, not proved. The scripts are `maple/10--14` (project) and `maple/20_paper_audit.mpl` (this paper).

2 Planar and spherical boundaries

Set $h = 1$. Place the particle at the origin and let the boundary be the plane (line) at distance $d \in [0, 1]$, the solid being the far side. λ then depends on d alone.

2.1 Three dimensions

Proposition 2.1 (Shell form). $\lambda_3(d) = \int_d^1 C_3 \widehat{W}(q) 2\pi q(q-d) dq$. [M]

Proof. The sphere of radius q about the particle meets the half space $z \geq d$ in a cap of height $q-d$ (for $q \geq d$) and area $2\pi q(q-d)$ (Archimedes); integrate shell by shell. $\square$

Wendland kernels. With $\widehat{W} = (1-q)^m P(q)$, $P = \sum_p c_p q^p$, put $F(m, p, x) = \int_0^x t^p (1-t)^m dt = \sum_{j=0}^m (-1)^j \binom{m}{j} \frac{x^{p+j+1}}{p+j+1}$. Then

$$\lambda_3(d) = 2\pi C_3 \sum_p c_p \left[F(m, p+2, 1) - F(m, p+2, d) - d(F(m, p+1, 1) - F(m, p+1, d)) \right], \quad (5)$$

a single polynomial in d of degree $m + \deg P + 3$, i.e. 8, 11, 14 for w2, w4, w6, with $\lambda_3(0) = \frac{1}{2}$, $\lambda_3(1) = 0$. [M]

Cubic spline. The knot at $q = \frac{1}{2}$ splits the integral (inner correction $-4(\frac{1}{2} - q)^3$ only for $q \leq \frac{1}{2}$):

$$d \in [0, \frac{1}{2}] : \quad \lambda_3 = \frac{1}{60} (192d^6 - 288d^5 + 160d^3 - 84d + 30), \quad (6)$$

$$d \in [\frac{1}{2}, 1] : \quad \lambda_3 = -\frac{8}{15} (2d^6 - 9d^5 + 15d^4 - 10d^3 + 3d - 1), \quad (7)$$

which agree at $d = \frac{1}{2}$ (value $\frac{1}{30}$).

[M] (Winchenbach et al. [2] give the cubic case.)

2.2 Two dimensions

Proposition 2.2 (Arc form). $\lambda_2(d) = \int_d^1 C_2 \widehat{W}(q) 2q \arccos\left(\frac{d}{q}\right) dq.$ [M]

Proof. The circle of radius $q \geq d$ about the particle meets the half plane $y \geq d$ where $q \cos \phi \geq d$, i.e. in the arc $|\phi| \leq \arccos(d/q)$ of length $2q \arccos(d/q)$. Integrate shell by shell. (Compare 3D: there the shell area is $2\pi q(q-d)$, polynomial in q ; here the inverse cosine makes the integrand non-polynomial.) $\square$

Proposition 2.3 (By-parts reduction). *Let $v_0(q) = \int_0^q t \widehat{W}(t) dt$, so that $2\pi C_2 v_0(1) = 1$. Then*

$$\lambda_2(d) = 2C_2 \left[v_0(1) \arccos d - d \int_d^1 \frac{v_0(q)}{q \sqrt{q^2 - d^2}} dq \right]. \quad (8)$$

In particular $\lambda_2(0) = \frac{1}{2}$, $\lambda_2(1) = 0$ and $\lambda_2'(d) = -2C_2 \int_d^1 \widehat{W}(q) q (q^2 - d^2)^{-1/2} dq$. [M]

Proof. Integrate Proposition 2.2 by parts with $u = \arccos(d/q)$ and $dv = 2C_2 q \widehat{W}(q) dq$, i.e. $v = 2C_2 v_0$. Since $\frac{d}{dq} \arccos(d/q) = d/(q \sqrt{q^2 - d^2})$, $u(d) = \arccos 1 = 0$ and $u(1) = \arccos d$,

$$\lambda_2 = [uv]_d^1 - \int_d^1 v u' dq = 2C_2 v_0(1) \arccos d - 2C_2 d \int_d^1 \frac{v_0(q)}{q \sqrt{q^2 - d^2}} dq.$$

At $d = 0$ the second term vanishes and $\lambda_2(0) = 2C_2 v_0(1) \cdot \frac{\pi}{2} = \frac{1}{2}$. For the derivative differentiate the arc form directly: the boundary term at $q = d$ vanishes because $\arccos 1 = 0$, and $\partial_d \arccos(d/q) = -(q^2 - d^2)^{-1/2}$. $\square$

Lemma 2.4 (The family $\mathcal{J}_m$). *Let $\mathcal{J}_m(d) = \int_d^1 q^m (q^2 - d^2)^{-1/2} dq$. Then $\mathcal{J}_0 = \ln \frac{1+\sqrt{1-d^2}}{d}$, $\mathcal{J}_1 = \sqrt{1-d^2}$ and*

$$m \mathcal{J}_m = \sqrt{1-d^2} + (m-1) d^2 \mathcal{J}_{m-2} \quad (m \geq 2). \quad (9)$$

Consequently, for odd m , $\mathcal{J}_m = \sqrt{1-d^2} \text{poly}(d^2)$; for even m , $\mathcal{J}_m = \sqrt{1-d^2} \text{poly}(d^2) + \text{poly}(d^2) \mathcal{J}_0$, where $\mathcal{J}_0$ contains $\ln d$ and $\ln(1 + \sqrt{1-d^2})$. [M]

Proof. $\mathcal{J}_0 = \text{arcosh}(1/d)$ and $\mathcal{J}_1$ are direct. For $m \geq 2$, using $\sqrt{q^2 - d^2} = (q^2 - d^2)/\sqrt{q^2 - d^2}$,

$$\frac{d}{dq} \left(q^{m-1} \sqrt{q^2 - d^2} \right) = (m-1) q^{m-2} \sqrt{q^2 - d^2} + \frac{q^m}{\sqrt{q^2 - d^2}} = \frac{m q^m - (m-1) d^2 q^{m-2}}{\sqrt{q^2 - d^2}}.$$

Integrate over $[d, 1]$: the left side gives $\sqrt{1-d^2}$ (the term at $q = d$ vanishes). The parity statements follow by induction from $\mathcal{J}_1$ and $\mathcal{J}_0$. For example $\mathcal{J}_3 = \frac{1}{3}(1 + 2d^2)\sqrt{1-d^2}$ and $\mathcal{J}_2 = \frac{1}{2}(\sqrt{1-d^2} + d^2 \mathcal{J}_0)$. $\square$

Remark 2.5. The project notes state that odd $\mathcal{J}_m$ lie in $\text{poly}(d^2)\{1, \sqrt{1-d^2}\}$; Eq. (9) shows the constant is never present (Appendix A).

Wendland. Because $\widehat{W}(q) = 1 + O(q^2)$, $v_0 = O(q^2)$ is a polynomial and $v_0(q)/q = \sum_{m \geq 1} a_m q^m$. Then (8) reads

$$\lambda_2(d) = 2C_2 \left[v_0(1) \arccos d - d \sum_{m \geq 1} a_m \mathcal{J}_m(d) \right], \quad (10)$$

that is $\arccos d$ plus $d [\text{poly} \sqrt{1-d^2} + \text{poly} \ln d + \text{poly} \ln(1 + \sqrt{1-d^2})]$. *Example (w2).* $\widehat{W} = 1 - 10q^2 + 20q^3 - 15q^4 + 4q^5$ gives $v_0 = \frac{q^2}{2} - \frac{5q^4}{2} + 4q^5 - \frac{5q^6}{2} + \frac{4q^7}{7}$, $v_0(1) = \frac{1}{14}$ (so $2C_2 v_0(1) = \frac{1}{\pi}$ with $C_2 = \frac{7}{\pi}$), and

$$\lambda_2^{w2}(d) = \frac{14}{\pi} \left[\frac{\arccos d}{14} - d \left(\frac{1}{2} \mathcal{J}_1 - \frac{5}{2} \mathcal{J}_3 + 4 \mathcal{J}_4 - \frac{5}{2} \mathcal{J}_5 + \frac{4}{7} \mathcal{J}_6 \right) \right] \quad [\text{M, 40 digits at } d = \frac{3}{10}].$$

Cubic. The knot splits the range. For $d \geq \frac{1}{2}$ only the outer piece $\widehat{W} = (1 - q)^3$ is present and the Wendland recipe applies. For $d < \frac{1}{2}$ the inner correction $-4(\frac{1}{2} - q)^3$ is a second block with upper limit $\frac{1}{2}$; its integrals end at $q = \frac{1}{2}$ instead of 1, which adds terms in $\arccos 2d$, $\sqrt{1 - 4d^2}$ and $\ln(1 + \sqrt{1 - 4d^2})$. At $d = \frac{1}{2}$ only the outer piece contributes, and

$$\lambda_2(\tfrac{1}{2}) = \frac{64\pi - 294\sqrt{3} + 249 \ln(2 + \sqrt{3})}{168\pi} \approx 0.03744 \quad [\text{M}, 40 \text{ digits}].$$

2.3 Solid ball in 3D

Solid ball of radius R , particle outside at distance d from the surface, hence $D = R + d$ from the centre.

Proposition 2.6 (Cap area). *The part of the sphere $|\mathbf{y}| = q$ (centre at the particle) that lies inside the ball has area $A(q) = \frac{\pi q}{D} (R^2 - D^2 + 2Dq - q^2)$ for $q \in (D - R, D + R)$, and 0 otherwise. Hence $\lambda_{\text{sph}}(R, d) = \int_d^{\min(1, 2R+d)} C_3 \widehat{W}(q) A(q) dq$. [M]*

Proof. A point of the sphere at polar angle θ from the direction to the ball centre has squared distance $q^2 + D^2 - 2Dq \cos \theta$ to that centre, so it lies in the ball iff $\cos \theta \geq (q^2 + D^2 - R^2)/(2Dq)$. This is a spherical cap of height $h_c = q(1 - \cos \theta_0) = q - (q^2 + D^2 - R^2)/(2D)$ and area $2\pi q h_c$ (Archimedes), which is the stated A . It vanishes at $q = D \mp R$ [M], where the sphere touches the ball from outside or from inside. Beyond $q = D + R$ the sphere encloses the ball, so $A = 0$. $\square$

Two branches arise: $2R + d \geq 1$ (the support truncates, upper limit 1) and $2R + d < 1$ (the ball lies entirely inside the support, upper limit $2R + d$). For Wendland kernels A is quadratic in q , so the F -basis of Eq. (5) closes both branches (with one more power of q) and $\lambda_{\text{sph}} = N(R, d)/(R + d)$ with N a polynomial; on branch 1, N is linear in R because $R^2 - D^2 = -(2Rd + d^2)$ is. Since $\lim_{R \rightarrow \infty} A = 2\pi q(q - d)$ [M], branch 1 recovers Eq. (5).

2.4 Consistency of the two routes

Planar closed forms are the exact oracles against which every later construction is compared. Their independence from the edge calculus of Section 3 is established in Proposition 3.9.

3 Edge calculus in two dimensions

This is the core of the construction. For a polygon T we reduce $\int_T W$, its gradient and all moments to one-dimensional integrals along the edges. For piecewise-polynomial W these are elementary. The plan is the table of Section 1.3: a value identity (Theorem 3.3), a gradient identity (Theorem 3.4) and a recursion for moments (Theorem 3.7); Proposition 3.1 supplies the one-dimensional integrals they need.

3.1 Geometry

Let T be a simple polygon, counter-clockwise. For the evaluation point $\mathbf{x}$ and an edge $e = (p_e, q_e)$ let $\mathbf{t}_e$ be the unit tangent, $\mathbf{n}_e$ the *outward* normal, and

$$\begin{aligned} z_e &= \mathbf{n}_e \cdot (p_e - \mathbf{x}), & s_0 &= \mathbf{t}_e \cdot (p_e - \mathbf{x}), & s_1 &= \mathbf{t}_e \cdot (q_e - \mathbf{x}), \\ \mathbf{y} &= z_e \mathbf{n}_e + s \mathbf{t}_e, & r &= \sqrt{s^2 + z_e^2} \quad \text{on the edge line.} \end{aligned} \quad (11)$$

So s is the position along the edge measured from the foot of the perpendicular from $\mathbf{x}$, and z_e is the distance to the edge line: $z_e > 0$ iff $\mathbf{x}$ lies on the inner side of the edge line. For a profile with cut radius r_c , the edge meets the support in the *chord* $s \in [\max(s_0, -\ell), \min(s_1, \ell)]$, $\ell = \sqrt{r_c^2 - z_e^2}$, empty if $|z_e| \geq r_c$ or the interval is empty.

3.2 Elementary primitives

With $r = \sqrt{s^2 + z^2}$ define $I_m(s) = \int_0^s r^m ds$ and $S_{j,m} = \int s^j r^m ds$.

Proposition 3.1 (Primitives). (a) $(m+1)I_m = s r^m + m z^2 I_{m-2}$ with $I_0 = s$, $I_{-1} = \text{asinh}(s/|z|)$, $z I_{-2} = \arctan(s/z)$. Hence for even $m = 2k \geq 0$, $I_m = \sum_i \binom{k}{i} z^{2k-2i} \frac{s^{2i+1}}{2i+1}$ (a polynomial), and for odd $m \geq 1$, $I_m = \frac{m!!}{(m+1)!!} z^{m+1} \text{asinh}(s/|z|) + s r \text{poly}(r^2, z^2)$.

(b) $\int s r^m ds = r^{m+2}/(m+2)$ for $m \neq -2$, and $\ln r$ for $m = -2$.

(c) Powers of s are reduced with $s^2 = r^2 - z^2$. For $j = 2p$ even, $S_{j,m} = \sum_{i=0}^p \binom{p}{i} (-z^2)^{p-i} I_{m+2i}$; for $j = 2p+1$ odd, $S_{j,m} = \sum_{i=0}^p \binom{p}{i} (-z^2)^{p-i} \frac{r^{m+2i+2}}{m+2i+2}$ (with $\ln r$ if $m+2i+2 = 0$). Note that $p = \lfloor j/2 \rfloor$ in both cases.

(d) For a chord $[lo, hi]$ and either sign of z : $\arctan(hi/z) - \arctan(lo/z) = \text{atan2}(z(hi-lo), z^2 + hi \cdot lo)$. [M, maple/10]

Proof. (a) Differentiating $s r^m$ gives $r^m + m s^2 r^{m-2} = (m+1)r^m - m z^2 r^{m-2}$, which is the recurrence; the closed forms follow by induction from I_0 and $I_1 = \frac{1}{2}(s r + z^2 \text{asinh}(s/|z|))$. (b) is the chain rule. (c) For $j = 2p$ expand $s^{2p} r^m = (r^2 - z^2)^p r^m$ binomially and integrate term by term; for $j = 2p+1$ write $s^{2p+1} r^m = s(r^2 - z^2)^p r^m$ and use (b). (d) is the tangent subtraction formula $\tan(a-b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$ with the denominator and numerator multiplied by $z^2 > 0$; the atan2 form selects the correct branch since the difference lies in $(-\pi, \pi)$ and has the sign of $z(hi-lo)$. All derivative identities were verified symbolically for $|m| \leq 12$, $j \leq 7$. $\square$

Remark 3.2. I_m is odd in s , but $\int s r^m ds$ is even; a chord spanning $s = 0$ therefore adds I_m contributions but subtracts equal contributions of the second kind. For $m < -2$ the downward recurrence divides by z^2 each step and is numerically unusable at small z ; only $m \geq 0$ is needed by the production path.

3.3 Truncated monomials

Any piecewise-polynomial radial kernel is a finite combination of *blocks* $f_{n,r_c}(r) = r^n \mathbb{K}[r \leq r_c]$: $W = \sum_j q_j(r) \mathbb{K}[r \leq r_j]$ with q_{last} the outer piece and $q_j = \text{piece}_j - \text{piece}_{j+1}$.

- Wendland: one block family with $r_c = 1$.
- Cubic: $W = C_2[(1-q)^3 - 4(\frac{1}{2}-q)^3 \mathbb{K}[q \leq \frac{1}{2}]]$, blocks at $r_c = 1$ and $r_c = \frac{1}{2}$.

[M]

3.4 Value identity

For a radial g supported on $r \leq r_c$ set $M_g(r) = \int_0^r t g(t) dt$ (so $M_g(r) = M_g(r_c)$ for $r \geq r_c$).

Theorem 3.3 (Value identity). Let $\mathbf{x} \notin \partial T$. Then

$$\int_T g(|\mathbf{x}' - \mathbf{x}|) d\mathbf{x}' = \mathbb{K}[\mathbf{x} \in T] 2\pi M_g(r_c) + \sum_{e \in \partial T} z_e \int_{\text{chord}_e} \frac{M_g(r) - M_g(r_c)}{r^2} ds. \quad (12)$$

For a normalised kernel $2\pi M_W(1) = 1$, so the first term is $\mathbb{K}[\mathbf{x} \in T]$.

Proof. Step 1: g is a divergence. Put $\mathbf{G}(\mathbf{y}) = \mathbf{y} M_g(r)/r^2$. Away from $\mathbf{y} = 0$, $\nabla \cdot \mathbf{G} = \frac{2M_g}{r^2} + r \frac{d}{dr} \frac{M_g}{r^2} = \frac{2M_g}{r^2} + \frac{M'_g}{r} - \frac{2M_g}{r^2} = \frac{M'_g}{r} = g$, since $M'_g = r g$.

Step 2: divergence theorem. If $\mathbf{x} \notin T$, $\mathbf{G}$ is smooth on T and $\int_T g = \oint_{\partial T} \mathbf{G} \cdot \mathbf{n} ds$. If $\mathbf{x} \in T$, apply

it on $T \setminus B_\varepsilon(\mathbf{x})$; the flux through the small circle (normal pointing towards $\mathbf{x}$) is $-\frac{M_g(\varepsilon)}{\varepsilon} \cdot 2\pi\varepsilon = -2\pi M_g(\varepsilon) \rightarrow 0$ since $M_g(\varepsilon) = O(\varepsilon^2)$. On an edge $\mathbf{n}_e \cdot \mathbf{y} = z_e$, so in both cases

$$\int_T g = \sum_e z_e \int_e \frac{M_g(r)}{r^2} ds \quad (\text{whole edges}). \quad (13)$$

Step 3: localise. Write $M_g = (M_g - M_g(r_c)) + M_g(r_c)$. The first part vanishes for $r \geq r_c$, so only the chord contributes. For the constant part, $z_e \int ds/r^2 = \arctan(s/z_e)|_{s_0}^{s_1}$ is the angle under which edge e is seen from $\mathbf{x}$, and $\sum_e$ of these is the total angle subtended by the closed curve ∂T , namely 2π if $\mathbf{x} \in T$ and 0 otherwise. $\square$

This is the identity behind all edge-based boundary methods; the 3D analogue replaces 2π by 4π and the angle by the solid angle (Section 8). In words: only the edges near $\mathbf{x}$ matter, plus a topological term that counts whether $\mathbf{x}$ is inside.

Block form. For $g = r^n \mathbb{K}[r \leq r_c]$ one has $M_g = \min(r, r_c)^{n+2}/(n+2)$ and $(M_g - M_g(r_c))/r^2 = (r^n - r_c^{n+2}r^{-2})/(n+2)$ on the chord, so by $z \int ds/r^2 = \arctan(s/z)$,

$$\int_T r^n \mathbb{K}[r \leq r_c] = \mathbb{K}[\mathbf{x} \in T] \frac{2\pi r_c^{n+2}}{n+2} + \sum_e \frac{1}{n+2} \left[z_e (I_n|_{lo}^{hi}) - r_c^{n+2} \Delta \arctan \right]. \quad (14)$$

For a multi-block kernel apply Eq. (14) to each block and sum. Collecting the blocks of cut radius r_j into one group with radial mass $M_j(r_j)$, the group contributes the indicator weight $w_j = 2\pi M_j(r_j)$ and the term $-\frac{w_j}{2\pi} \Delta \arctan_j$ (the angle taken on the chord of radius r_j). For the cubic spline $w_j = \frac{8}{7}$ ($r_j = 1$) and $-\frac{1}{7}$ ($r_j = \frac{1}{2}$); the *negative* weight is correct and $\sum_j w_j = 1$. [M]

3.5 Degenerate placements

$\lambda_T(\mathbf{x}) = \int_T W$ is continuous in $\mathbf{x}$ (W is bounded), but the two parts of (12) are not. Take $\mathbf{x}$ crossing the line of an edge e at a point projecting into the edge interior, from outside to inside, so z_e changes sign from $-$ to $+$. The indicator jumps by $+2\pi M_W(1) = +1$. In the chord term of e the integrand $z_e (M_W(r) - M_W(1))/r^2$ concentrates at $s = 0$ as $|z_e| \rightarrow 0$, where $M_W(0) = 0$: it tends to $-M_W(1)\pi \operatorname{sgn} z_e$, i.e. it jumps by $-2\pi M_W(1) = -1$. The two jumps cancel; all other edges are continuous. Therefore the value *on* the boundary, obtained by continuity, uses the average of the two one-sided limits of each part: *$\mathbf{x}$ in the open interior of an edge: indicator $\frac{1}{2}$, that edge contributes 0 to the chord terms (and to every arctan of Eq. (14)); $\mathbf{x}$ at a vertex: indicator = interior angle/ 2π ; an edge with $z_e = 0$ contributes 0.* (At a vertex, the two incident edges each contribute $-M_W(1)(\pi - \frac{\alpha}{2})$ in the limit from the bisector, for a total $-M_W(1)(2\pi - \alpha)$, and $2\pi M_W(1) - M_W(1)(2\pi - \alpha) = M_W(1)\alpha$ is exactly the indicator $\alpha/2\pi$ times $2\pi M_W(1)$.) These conventions must use the *same* orientation predicate for the sign of z_e and for the indicator. [N]

Worked check: a flat wall. For a half-plane solid at distance d (single edge, $z = -d$, indicator 0), Eq. (12) gives $\lambda(d) = -d \int_{-\ell}^{\ell} \frac{M_W(r) - M_W(1)}{r^2} ds$. As $d \rightarrow 0$ the factor d kills the $M_W(r) \sim r^2$ part (the integrand is bounded) and what remains is $+d M_W(1) \int_{-\ell}^{\ell} ds/r^2 = 2M_W(1) \arctan(\ell/d) \rightarrow \pi M_W(1) = \frac{1}{2}$, the known value $\lambda(0) = \frac{1}{2}$ of Proposition 2.3.

3.6 Gradient identity

Theorem 3.4 (Gradient identity). *For g bounded, piecewise continuous with jumps on circles about $\mathbf{x}$ that meet ∂T transversally,*

$$\partial_{x_j} \int_T g(|\mathbf{x}' - \mathbf{x}|) d\mathbf{x}' = - \sum_e n_{e,j} \int_{\text{chord}_e} g(r) ds.$$

In particular, $\nabla_{\mathbf{x}}\lambda = -\sum_e \mathbf{n}_e \int_{\text{chord}_e} W \, ds$ without any indicator term, and the identity holds block by block.

Proof. $\int_T g(\mathbf{x}' - \mathbf{x}) d\mathbf{x}' = \int_{T-\mathbf{x}} g(\mathbf{y}) d\mathbf{y}$, so moving $\mathbf{x}$ by $\delta \mathbf{e}_j$ translates the domain by $-\delta \mathbf{e}_j$. A domain translated by $\mathbf{v}$ gains $\oint (\mathbf{v} \cdot \mathbf{n}) g \, ds + o(|\mathbf{v}|)$ (Reynolds transport; g is continuous across $\partial T - \mathbf{x}$ a.e.), here $-\delta \oint n_j g \, ds$. Only the chords carry $g \neq 0$. $\square$

Sign check: moving $\mathbf{x}$ towards the solid, i.e. against $\mathbf{n}_e$ (which points out of the solid), increases λ ; the formula gives $\nabla \lambda \parallel -\mathbf{n}_e$.

Corollary 3.5 (First moment). *With $\Psi(r) = \int_r^1 tW \, dt$ (so $\mathbf{y}W = -\nabla \Psi$),*

$$\int_T \mathbf{y}W d\mathbf{x}' = -\sum_e \mathbf{n}_e \int_{\text{chord}_e} \Psi \, ds. \quad [N]$$

This is the case $|\alpha| = 1$ of Theorem 3.7 with $\Psi = -\Phi[W]$.

Corollary 3.6. *Gradients of moments are edge-local: $\partial_{x_j} m_\alpha = -\sum_e n_{e,j} \int_{\text{chord}_e} \mathbf{y}^\alpha W \, ds$. (The proof of Theorem 3.4 translates the domain $T - \mathbf{x}$ and so differentiates both $\mathbf{y}^\alpha$ and W .) If T covers the support the sum vanishes, as it must.*

3.7 Compact-potential recursion for moments

Let f be radial and supported on $r \leq r_c$ and $\Phi[f](r) = -\int_r^{r_c} t f(t) \, dt = M_f(r) - M_f(r_c)$. Then Φ is compactly supported, continuous, and $\partial_i \Phi = y_i f$ (since $\Phi'(r) = rf$ and $\partial_i r = y_i/r$).

Theorem 3.7 (Recursion). *For $\alpha = \beta + \mathbf{e}_i$,*

$$\int_T \mathbf{y}^\alpha f \, d\mathbf{x}' = \sum_e n_{e,i} \int_{\text{chord}_e} \mathbf{y}^\beta \Phi \, ds - \beta_i \int_T \mathbf{y}^{\beta-\mathbf{e}_i} \Phi \, d\mathbf{x}'. \quad (15)$$

Proof. $\partial_i(\mathbf{y}^\beta \Phi) = \beta_i \mathbf{y}^{\beta-\mathbf{e}_i} \Phi + \mathbf{y}^\beta y_i f$; integrate over T and use the divergence theorem on the left. Φ is Lipschitz, so no jump term arises even if f jumps at r_c . $[M, k \leq 4] \quad \square$

The degree drops by 2 per step. For odd k the recursion ends in pure edge integrals. For even k it ends in a value-type integral of an iterated potential, evaluated by Theorem 3.3 with M_g of that potential — which is why Theorem 3.3 is stated for general g with $2\pi M_g(r_c) \neq 1$. For a block $f = r^n \mathbb{K}[r \leq r_c]$, $\Phi = \frac{r^{n+2} - r_c^{n+2}}{n+2} \mathbb{K}[r \leq r_c]$ again lies in the block family, so only I_m and $\int s r^m ds$ with $m \geq 0$ occur.

Numerical confirmation. For a non-convex hexagon, all four kernels, particle inside, outside and near an edge, all 28 multi-indices with $|\alpha| \leq 6$ agree with an independent ray quadrature of the area integral to relative $3 \cdot 10^{-11}$ (`scripts/derivation_checks/paper_edge_identities_k6.py`); this covers the range $k \leq 6$ needed by Section 6 beyond the symbolic check. The edge form of $\partial_{\mathbf{x}} m_\alpha$ (Corollary 3.6) agrees with central differences to the finite-difference error ($\leq 10^{-5}$) and $g_\alpha = \partial_{\mathbf{x}} m_\alpha + \sum_j \alpha_j \mathbf{e}_j m_{\alpha-\mathbf{e}_j}$ (Proposition 4.1) with a direct ray quadrature of $\mathbf{y}^\alpha \nabla_{\mathbf{x}} W$ to $4 \cdot 10^{-12}$. $[N]$

Example ($k = 2$). For $\alpha = 2\mathbf{e}_1$ take $\beta = \mathbf{e}_1$, $i = 1$: $\int_T y_1^2 f = \sum_e n_{e,1} \int_{\text{chord}_e} y_1 \Phi \, ds - \int_T \Phi$, where on the chord $y_1 = z_e n_{e,1} + s t_{e,1}$ is linear in s , and $\int_T \Phi$ is a value integral with $M_\Phi(r_c) \neq \frac{1}{2\pi}$ in general.

Remark 3.8 (Direct form (b) fails on an edge line). An alternative with $\mathbb{K}[\mathbf{x} \in T] \oint \omega^\alpha r_c^{n+2+k} / (n+2+k) + \sum_e \frac{z_e}{n+2+k} \int (\dots) r^{-(2+k)}$ is correct for $z_e \neq 0$ but $0 \cdot \infty$ at $z_e = 0$, with a direction-dependent limit: the average convention of the previous subsection is *wrong* for it when $k \geq 1$ ($\int_{\text{half circle}} \sin \theta \neq \frac{1}{2} \oint \sin \theta$). The recursion (15) has no such problem. $[N]$

3.8 Half plane and the planar closed form

Proposition 3.9 (Consistency with Section 2.2). *For T a half plane (a single infinite edge, $z = -d$, indicator 0, chord $[-\ell, \ell]$, $\ell = \sqrt{1-d^2}$), Eq. (12) equals $\lambda_2(d)$ for every kernel.*

Proof. Both sides vanish at $d = 1$, so it suffices to compare derivatives in d . For the edge side, Theorem 3.4 (moving $\mathbf{x}$ away from the wall by dd is a displacement along $\mathbf{n}$) gives $\partial_d \lambda = -\int_{-\ell}^{\ell} W(\sqrt{s^2 + d^2}) ds = -2 \int_d^1 W(q) q (q^2 - d^2)^{-1/2} dq$ (substituting $q = \sqrt{s^2 + d^2}$, $ds = q dq/s$), which is $\lambda'_2(d)$ by Proposition 2.3. $\square$

The two derivations are independent: the shell calculus integrates arcs, the edge calculus uses the divergence theorem. Note what is and is not tested: the proof compares the *gradient* identity with the arc form; the value identity itself enters through the single infinite edge (whose total angle $-\pi$ is closed by the arc at infinity, flux $+\pi M_W(1)$) and is confirmed by the check at $d \rightarrow 0$ above and numerically for all four kernels at $d \in \{0, 10^{-12}, \dots, 1 - 10^{-12}\}$ and symbolically [M, N].

4 Fields on elements: FEM nodal weights

Fluid–solid interactions weight a field A living on the solid. On a triangle T let $A(\mathbf{x}') = \sum_i A_i N_i(\mathbf{x}')$ with Lagrange basis N_i of degree p .

Proposition 4.1 (Nodal weights). *Expand $N_i(\mathbf{x} + \mathbf{y}) = \sum_{|\alpha| \leq p} B_{\alpha i}(\mathbf{x}) \mathbf{y}^\alpha$, $B_{\alpha i} = \partial^\alpha N_i(\mathbf{x})/\alpha!$ (exact, finite). Then*

$$\int_T A W d\mathbf{x}' = \sum_i A_i w_i, \quad w_i = \sum_{\alpha} B_{\alpha i} m_{\alpha}, \quad (16)$$

$$\int_T A \nabla_{\mathbf{x}} W d\mathbf{x}' = \sum_i A_i G_i, \quad G_i = \sum_{\alpha} B_{\alpha i} g_{\alpha}, \quad (17)$$

with $g_{\alpha} = \partial_{\mathbf{x}} m_{\alpha} + \sum_j \alpha_j \mathbf{e}_j m_{\alpha - \mathbf{e}_j}$ (the sum runs over the components j ; $m_{\alpha - \mathbf{e}_j} = 0$ if $\alpha_j = 0$), where $\partial_{\mathbf{x}} m_{\alpha}$ is given by Corollary 3.6.

Proof. The first two are linearity. For g_{α} : $m_{\alpha} = \int_T \mathbf{y}^{\alpha} W(|\mathbf{y}|)$ with $\mathbf{y} = \mathbf{x}' - \mathbf{x}$, so $\partial_{x_j} m_{\alpha} = -\alpha_j m_{\alpha - \mathbf{e}_j} + g_{\alpha, j}$, because ∂_{x_j} acts on both $\mathbf{y}^{\alpha}$ and on W , whereas g_{α} differentiates W only. $\square$

Hence the weights need exactly the moments m_{α} , $|\alpha| \leq p$ ($p \leq 3$ gives $k \leq 3$ for w_i , and $k \leq 3$ edge integrals for G_i), computed by Theorem 3.7. [N]

Special cases. $p = 0$: $w = m_0$. $p = 1$: N_i is the barycentric coordinate, $N_i(\mathbf{x} + \mathbf{y}) = N_i(\mathbf{x}) + \nabla N_i \cdot \mathbf{y}$, so $w_i = N_i(\mathbf{x}) m_0 + \nabla N_i \cdot \mathbf{m}_1$. Partition of unity $\sum_i w_i = \int_T W$ follows from $\sum_i N_i \equiv 1$. If T covers the support, the weights reproduce polynomial fields of degree $\leq p$ exactly ($\int A W$ and $\int A \nabla_{\mathbf{x}} W = \int \nabla A W$). [N]

Conditioning. The formulas are exact; their floating-point behaviour is not. Three exact reformulations remove the loss: (1) for polygons inside the support one may use the potential $\tilde{\Phi} = \int_0^r t f$ in (15), because it differs from Φ by the constant $C = M_f(r_c)$ (Theorem 3.3), whose total contribution $C(\oint n_i \mathbf{y}^{\beta} ds - \beta_i \int_T \mathbf{y}^{\beta - \mathbf{e}_i})$ vanishes by the divergence theorem; (2) the same for g_{α} , which can be written $g_{\alpha, j} = -\int_T \mathbf{y}^{\alpha + \mathbf{e}_j} L$ with $L = W'/r$ (because $\partial_{x_j} W = -L y_j$); the potential of L is $\Phi[L] = W$, and the shifted potential $\tilde{\Phi} = W - W(0)$ plays the role of $\tilde{\Phi}$ in (1); (3) for far elements the integrand $N_i W$ is smooth and Gauss quadrature converges spectrally. The practical error tables are in the project notes and tests and are not reproduced here. [N]

5 Closed boundaries (data on the boundary only)

A solid bounded by a closed polyline is a simple polygon S . Theorems 3.3, 3.4 and 3.7 hold for S unchanged, using only ∂S and the indicator $\mathbb{K}[\mathbf{x} \in S]$ (the crossing number, computed from the same z_e as the edge terms). For a triangulation of S the interior edges cancel pairwise (each appears twice with opposite $\mathbf{n}_e$). A field known only on ∂S enters through a polynomial extension of degree q (moments $k \leq q$).

5.1 Polygonal approximation of a curved boundary

Let S be an inscribed (or circumscribed) regular N -gon of a disk of radius R , $a = \pi/N$. In a polar angle $\varphi \in [-a, a]$ about an edge normal (centre of the disk at the origin), the edge sits at radius $\rho_S(\varphi) = R \cos a / \cos \varphi$ (inscribed) or $R / \cos \varphi$ (circumscribed), so the radial deviation from the circle is

$$\delta_{\text{in}}(\varphi) = R \left(\frac{\cos a}{\cos \varphi} - 1 \right) = \frac{R}{2}(\varphi^2 - a^2) + O(Ra^4), \quad \delta_{\text{circ}}(\varphi) = R \left(\frac{1}{\cos \varphi} - 1 \right) = \frac{R}{2}\varphi^2 + O(Ra^4),$$

with means over $[-a, a]$ of $-Ra^2/3$ and $+Ra^2/6$ (ratio $-2 : 1$). The same ratio holds for the area deficit and excess: $\frac{1}{2}NR^2 \sin 2a = \pi R^2(1 - \frac{2}{3}a^2 + \dots)$ and $NR^2 \tan a = \pi R^2(1 + \frac{1}{3}a^2 + \dots)$.

Proposition 5.1. *Let $\epsilon = \frac{Ra^2}{6} \int_0^{2\pi} w \, d\phi$ with $w(\phi) = RW(|\mathbf{x}'(R, \phi) - \mathbf{x}|)$ (the kernel on the circle). If $W \in C^2$ then $\lambda_{\text{in}} - \lambda = -2\epsilon + O(N^{-4})$ and $\lambda_{\text{circ}} - \lambda = +\epsilon + O(N^{-4})$, so $\frac{1}{3}\lambda_{\text{in}} + \frac{2}{3}\lambda_{\text{circ}}$ is accurate to $O(N^{-4})$.* [A, N]

Argument. In polar coordinates (ρ, ϕ) about the disk centre, S and the disk differ in the thin signed ring between $\rho = R$ and $\rho = R + \delta(\phi)$:

$$\lambda_S - \lambda = \int_0^{2\pi} \int_R^{R+\delta} W \rho \, d\rho \, d\phi = \int_0^{2\pi} w(\phi) \delta(\phi) \, d\phi + O(R^2 a^4).$$

The deviation δ is $2a$ -periodic with mean $\bar{\delta}$. Split $\delta = \bar{\delta} + (\delta - \bar{\delta})$. The oscillating part has zero mean on every period, so two integrations by parts against the 2π -periodic weight w (with w'' integrable, which needs $W \in C^2$) give $O(a \cdot a \cdot Ra^2) = O(a^4)$. Hence $\lambda_S - \lambda = \bar{\delta} \int w \, d\phi + O(a^4)$ with $\bar{\delta}_{\text{in}} = -Ra^2/3 + O(a^4)$, $\bar{\delta}_{\text{circ}} = Ra^2/6 + O(a^4)$. $\square$

Numerically (w4, $R = \frac{3}{2}$, $d = 0.3$, particle on an edge normal; `scripts/derivation_checks/paper_ngon_orders.py`) the errors of both polygons fall by 4 per doubling of N and their ratio tends to -2 . The combination divided by a^4 is 0.352, 0.374, 0.377, 0.377, 0.377 for $N = 24, 32, 48, 64, 96$, i.e. a clean $O(N^{-4})$ with coefficient ≈ 0.377 (1M and 4M rays agree to all digits shown). At small N the next term matters: the combination changes sign between $N = 12$ ($-0.35a^4$) and $N = 24$, passing through zero near $N = 16$, which is why a measurement of the order from $N = 8, 16, 32$ alone looked erratic (the earlier project value was ≥ 3.3). [N]

5.2 Limits

Self-intersecting or overlapping solids double-count (the indicator is a crossing number per solid) and are not covered. The polygon error $O(N^{-2})$ is exactly the regime where the curvature expansion of Section 6 is cheaper than adding edges.

6 Curvature expansion of the value integral

For smooth boundaries with radius of curvature large compared to h we expand λ in κ , the curvature at the closest boundary point, instead of adding edges. Set $h = 1$ and take $\kappa > 0$ for

a locally convex solid (a disk of radius R has $\kappa = 1/R$) and $\kappa < 0$ for a concave one (the solid is the complement of a disk). $\varphi(\sigma) = W(\sqrt{\sigma})$, $\sigma = |\mathbf{y}|^2$, extended by 0 for $\sigma > 1$. The idea: in the coordinates (arclength t of the closest boundary point, depth s) the curved solid is a flat strip with the weight $1 - \kappa s$, and the squared distance to the particle differs from the flat one by a power series in κ ; Taylor-expand φ in it.

6.1 Statement

Theorem 6.1 (Second-order closest-point expansion). *For a disk (more generally, a boundary whose curvature is constant over the support) and a particle at distance $d > 0$ outside,*

$$\lambda(d, \kappa) = F_0(d) + \kappa F_1(d) + \kappa^2 F_2(d) + O(\kappa^3), \quad (18)$$

where, with $y_1 = t$, $y_2 = d + s$ and integrals over the half plane $\{y_2 \geq d\}$,

$$F_0 = \int \varphi, \quad F_1 = \int \left[\varphi' y_1^2 (2d - y_2) - (y_2 - d) \varphi \right], \quad (19)$$

$$F_2 = \int \left[\varphi' (-d(y_2 - d)y_1^2 - \frac{y_1^4}{12}) + \frac{1}{2} \varphi'' y_1^4 (2d - y_2)^2 - (y_2 - d) \varphi' y_1^2 (2d - y_2) \right]. \quad (20)$$

$F_0 = \lambda_2$ of Section 2.2. The gradient is $\nabla_{\mathbf{x}} \lambda = \partial_d \lambda \hat{\mathbf{n}}$ (outward normal) in the constant-curvature case.

Proof. Step 1 (coordinates). Write a solid point as $\mathbf{x}' = c(t) - s \mathbf{n}(t)$, s the depth into the solid and t arclength of the closest boundary point. For the disk the polar area element $\rho' d\rho' d\theta$ with $\rho' = R - s$, $\theta = \kappa t$ gives $d\mathbf{x}' = (1 - \kappa s) dt ds$. The same holds for $\kappa < 0$ ($\rho' = |R| + s$).

Step 2 (distance). The particle is at distance $R + d$ from the centre, so by the law of cosines

$$\begin{aligned} |\mathbf{y}|^2 &= (d + s)^2 + (1 + \kappa d)(1 - \kappa s) \frac{2(1 - \cos \kappa t)}{\kappa^2} \\ &= \underbrace{(d + s)^2 + t^2}_{\sigma_0} + \kappa t^2 (d - s) + \kappa^2 \left(-d s t^2 - \frac{t^4}{12} \right) + O(\kappa^3), \end{aligned}$$

using $2(1 - \cos \kappa t)/\kappa^2 = t^2 - \kappa^2 t^4/12 + O(\kappa^4)$ and $(1 + \kappa d)(1 - \kappa s) = 1 + \kappa(d - s) - \kappa^2 ds$. *Step 3 (Taylor).* With $\delta_1 = t^2(d - s)$ and $\delta_2 = -d s t^2 - t^4/12$, expand $(1 - \kappa s) \varphi(\sigma_0 + \kappa \delta_1 + \kappa^2 \delta_2)$ to $O(\kappa^2)$:

$$\kappa^1 : \varphi' \delta_1 - s \varphi, \quad \kappa^2 : \varphi' \delta_2 + \frac{1}{2} \varphi'' \delta_1^2 - s \varphi' \delta_1.$$

These are exactly the integrands of F_1 and F_2 (with $d - s = 2d - y_2$ and $s = y_2 - d$).

Step 4 (remainder). Step 3 is the Taylor expansion of φ about σ_0 with increment $\delta\sigma = \kappa \delta_1 + \kappa^2 \delta_2 + O(\kappa^3) = O(\kappa)$ on the (bounded) support. For $d \geq d_0 > 0$ one has $\sigma_0 \geq d_0^2$, so $\sigma \geq d_0^2 - O(\kappa)$ stays away from $\sigma = 0$, where φ'' is singular for some kernels ($\varphi'' \sim 15/r$ for w2 and $9/4/r$ for the cubic; it is bounded for w4 and w6). Away from 0, φ'' is continuous and piecewise C^1 , hence Lipschitz, for all four kernels: the cubic has W''' jumping at $q = 1$ and $q = \frac{1}{2}$, but φ'' involves only W , W' and W'' , which are continuous. The Taylor remainder after the quadratic term is therefore bounded by $\frac{1}{6} \text{Lip}(\varphi'') |\delta\sigma|^3 = O(\kappa^3)$, uniformly on $d \geq d_0$. (The constant degrades as $d_0 \rightarrow 0$ because of the singularity; this is why the limit $d \rightarrow 0$ needs separate treatment.) $\square$

Steps 1–3 are verified symbolically

[M, maple/14 and maple/20].

6.2 Evaluation

Each integral is a sum of half-plane *moments* $m_{ij}[f] = \int_{y_2 \geq d} y_1^i y_2^j f(r)$ of the exact piecewise polynomial profiles W , $\varphi' = W'/(2r)$ and $\varphi'' = (W'/r)'/(4r)$. As polynomials in r these reach

down to r^{-1} for w2 (where $\varphi'' = 15/r + \dots$), to r^0 for w4 and w6, and for the cubic to r^{-3} in each block (the singular parts of the two blocks cancel to a net $\frac{9}{4}/r$). Each moment is a single edge instance of Theorem 3.7 (here $i + j \leq 6$; the recursion is general, its symbolic check in the project covers $k \leq 4$): no new quadrature. In practice $F_k(d)$ are tabulated once as one-dimensional functions.

6.3 Accuracy

The measured errors (Wendland C4, $d = 0.3h$) follow the predicted orders:

κh	exact λ	planar F_0	$+\kappa F_1$	$+\kappa^2 F_2$
1/2	0.09017	$9.0 \cdot 10^{-3}$	$9.9 \cdot 10^{-4}$	$1.4 \cdot 10^{-4}$
1/4	0.09444	$4.7 \cdot 10^{-3}$	$2.6 \cdot 10^{-4}$	$1.8 \cdot 10^{-5}$
1/8	0.09674	$2.4 \cdot 10^{-3}$	$6.8 \cdot 10^{-5}$	$2.3 \cdot 10^{-6}$
1/16	0.09794	$1.2 \cdot 10^{-3}$	$1.7 \cdot 10^{-5}$	$3.0 \cdot 10^{-7}$

(errors against the exact disk, [N]; recomputed independently with a shell quadrature of the exact disk and the half-plane integrals $F_0, F_1, F_2 = 0.09917, -0.01999, 0.004497$, all entries reproduced); successive halving of κ divides the three error columns by $\approx 2, 4, 8$.

6.4 General curves

Remark 6.2 (Variable curvature). Let the boundary have $\kappa(t) = \kappa + \kappa' t + \frac{1}{2} \kappa'' t^2 + \dots$ about the closest point of the particle. Count orders by dimension: $\kappa \sim h^{-1}$, $\kappa' \sim h^{-2}$, $\kappa'' \sim h^{-3}$. Thus κ' could enter Eq. (18) at order 2, next to κ^2 . It does not: the particle lies on the normal through the closest point, so the reflection $t \mapsto -t$ maps the particle to itself, sends $\kappa' \mapsto -\kappa'$ and leaves λ unchanged. Hence λ is even in κ' and has no term linear in κ' ; the first correction beyond Eq. (18) is κ'' (and κ'^2 at order 4), both of order ≥ 3 . [N] The *gradient* is different. With $\kappa(\mathbf{x})$ the curvature at the closest boundary point, $\nabla \lambda = \partial_d \lambda \hat{\mathbf{n}} + F_1 \nabla \kappa + \dots$ (κ is constant along normals), so there is a tangential part $\kappa' F_1 \hat{\mathbf{t}}$ of order 2. The gradient formula of Theorem 6.1 therefore has an $O(\kappa^2)$ tangential error unless $\kappa' F_1$ is added; for constant curvature it is exact to the order of the value expansion. At the next order the tangential component is $\kappa' [F_1 + \kappa(2F_2 - dF_1)]$: the factor $2F_2$ is $\partial_\kappa(\kappa^2 F_2)$, and $-dF_1$ comes from $\partial_x \kappa = \kappa'/(1 + \kappa d)$ (a tangential displacement ε of the particle moves the closest point by $\varepsilon/(1 + \kappa d)$). [N] Numerically (w4, $d = 0.3$,

boundary $y = -\frac{\kappa}{2}x^2 - \frac{\kappa'}{6}x^3$, `scripts/derivation_checks/paper_curvature_variable.py`): $\lambda(\kappa, \kappa') - \lambda(\kappa, 0)$ falls by 4 when κ' is halved (even in κ'); the parabola with $\kappa' = \kappa^2$ is reproduced by $F_0 + \kappa F_1 + \kappa^2 F_2$ with error falling by 8.0 per halving of κ ; the tangential gradient at $(\kappa, \kappa') = (\frac{1}{8}, \frac{1}{64})$ is $-2.840 \cdot 10^{-4}$ against $-3.123 \cdot 10^{-4}$ for $\kappa' F_1$ alone and $-2.830 \cdot 10^{-4}$ with the correction.

Remark 6.3 (Validity). The expansion is asymptotic in κh (usable for $\kappa h \lesssim \frac{1}{4}$ at 10^{-5}), requires reach $\geq h$ on the solid side, a single boundary sheet in the support, and $R > h - d$ so the support does not engulf the disk; the limit $d \rightarrow 0$ needs the limit form of the negative-power primitives.

7 Slender and thin obstacles

When a body is thin compared to h (fibres, strands) its volume integral reduces to a line integral with a short correction series.

Lemma 7.1 (Pizzetti mean-value series). *For f real-analytic on $\overline{B_a(c)} \subset \mathbb{R}^2$,*

$$\frac{1}{\pi a^2} \int_{B_a(c)} f = \sum_{k \geq 0} \frac{(a^2/4)^k}{k! (k+1)!} \Delta^k f(c).$$

Proof. Let $u(\rho) = \frac{1}{2\pi} \oint f(c + \rho\omega) d\omega$ be the circle mean. Averaging commutes with Δ , and for a radial function the 2D Laplacian is $u'' + u'/\rho$, so $u'' + u'/\rho = \Delta f(\rho)$, the circle mean of Δf (Darboux). Insert $u = \sum_k u_k \rho^{2k}$ with $u_0 = f(c)$: $(2k)^2 u_k = [\rho^{2k-2}] \Delta f = u_{k-1}[\Delta f]$, so by induction $u_k = \Delta^k f(c)/(4^k (k!)^2)$ (this is Pizzetti's formula [6]). Average over $\rho \in [0, a]$ with weight $2\rho/a^2$: $\int_0^a 2\rho^{2k+1} d\rho/a^2 = a^{2k}/(k+1)$, which turns $\frac{1}{4^k (k!)^2}$ into $\frac{1}{4^k k! (k+1)!}$. $\square$

Proposition 7.2 (Disk in 2D, strand in 3D). (a) A disk of radius a at distance D from the particle, with W smooth on $[D - a, D + a]$: $\int_{B_a} W = \pi a^2 \sum_{k \leq K} \frac{(a^2/4)^k}{k! (k+1)!} \Delta^k W(D) + O(a^{2K+4})$. (b) Straight 3D strand of radius a along a segment: with $\Delta_\perp$ the Laplacian in the cross-section,

$$\int_{\text{strand}} W \simeq \pi a^2 \int_{\text{axis}} \left[W + \frac{a^2}{8} \Delta_\perp W + \frac{a^4}{192} \Delta_\perp^2 W + \dots \right] ds,$$

the coefficients $\frac{1}{4 \cdot 1! \cdot 2!}$ and $\frac{1}{16 \cdot 2! \cdot 3!}$ of Lemma 7.1.

(c) 2D strip of half width a : $\int_{-a}^a f(z + u) du = \sum_k \frac{2a^{2k+1}}{(2k+1)!} f^{(2k)}(z)$.

Proof. (a),(b) apply Lemma 7.1 (in the cross-section for (b)); the first neglected term is $\pi a^2 \cdot a^{2K+2}$. (c) is the Taylor series integrated over u . The leading term of (b) is the line integral of W along the axis chord, $\pi a^2 \int_{\text{chord}} W ds$ with z the distance of the particle to the axis. [M for the coefficients and for (c); N for the orders 4, 6, 8 of (a), the 2D case: error ratios 15.7, 63.6, 256.1 per halving of a for $K = 0, 1, 2$ (w4, $D = 0.5$, paper_curvature_variable.py)] $\square$

Remark 7.3 (Regularity). Wendland kernels have no linear term, so $\Delta_\perp W$ is regular on the axis; but C^2 Wendland has an r^3 term, hence $\Delta_\perp^2 W \sim 1/r$ is singular where the axis passes through the particle, and the series is asymptotic and breaks at kernel knots or the support rim. Where it fails the polygon of Section 5 is the fallback; no gap exists in 2D since the strip is exactly $\lambda_2(d - a) - \lambda_2(d + a)$.

Ball in 3D. For a ball of radius ρ at centre distance D the *shell theorem* reads $\int_{S^2} f(|\mathbf{x} - \mathbf{c} - \rho\omega|) d\omega = \frac{2\pi}{D\rho} \int_{|D-\rho|}^{D+\rho} r f(r) dr$ [M], consistent with Proposition 2.6.

Curved centre line. A first-order correction $\propto \kappa_c a^2$ is expected (the tube volume is independent of κ_c but the kernel weighting is not symmetric); not derived. [P]

8 Extension to three dimensions

Theorem 8.1 (Face identity for a polyhedron T). For g radial with support R and $M_3(r) = \int_0^r t^2 g dt$,

$$\int_T g = \mathbb{K}[\mathbf{x} \in T] 4\pi M_3(R) + \sum_{f \in \partial T} z_f \int_f \frac{M_3(r) - M_3(R)}{r^3} dA,$$

$z_f = \mathbf{n}_f \cdot (p_f - \mathbf{x})$. For a normalised kernel $4\pi M_3(1) = 1$.

Proof. $\nabla \cdot (\mathbf{y} M_3/r^3) = 3M_3/r^3 + r(M_3'/r^3 - 3M_3/r^4) = M_3'/r^2 = g$ [M]. Apply the divergence theorem on $T \setminus B_\varepsilon(\mathbf{x})$ (small-sphere flux $-4\pi M_3(\varepsilon) \rightarrow 0$). For the constant part, $\sum_f z_f \int_f dA/r^3$ is the total solid angle subtended by ∂T ($= 4\pi$ inside, 0 outside). $\square$

The integrand on each face is compact: it is a 2D value-type integral in the face plane with profile $z_f(M_3(r) - M_3(R))/r^3$, $r^2 = \rho^2 + z_f^2$, to which Theorem 3.3 (in the plane, with the foot of $\mathbf{x}$) applies, leaving edge integrals of the form $\int r^m/(s^2 + w_e^2) ds$ with $r^2 = s^2 + w_e^2 + z_f^2$, which are elementary (solid-angle-type $\arctan(sz/(wr))$), cf. Van Oosterom–Strackee [5]).

Status. The identity above is proved; the *evaluation* (edge primitives in 3D, moment recursion in the face plane, comparison with the planar λ_3 and the sphere closed form, polyhedral potentials) is a plan. [P] The divergence step is verified symbolically; nothing else in this section has been numerically checked.

9 Wall operators built on the edge calculus

The same edge integrals provide terms of the SPH momentum equation. Here $\mathbf{y} = \mathbf{x}' - \mathbf{x}$ and $\mathbf{x}$ is a fluid particle, the solid half plane being locally the wall. Quantities are 2D. The solid is modelled as a continuum of mass m_{wall} per unit area, measured in units of the reference density ρ_0 , so a sum $\sum_j V_j(\cdot)$ over solid particles of volume $V_j = m_j/\rho_j$ becomes $\frac{m_{\text{wall}}}{\rho_i} \int_S(\cdot) d\mathbf{x}'$ with ρ_i the density of the fluid particle (also in units of ρ_0).

9.1 The wall Laplacian

Proposition 9.1 (Green form). *For the solid polygon S define $\lambda_\Delta(\mathbf{x}) := \int_S \nabla_{\mathbf{x}}^2 W(|\mathbf{x} - \mathbf{x}'|) d\mathbf{x}'$. Then*

$$\lambda_\Delta = \sum_e z_e \int_{\text{chord}_e} \frac{W'(r)}{r} ds. \quad (21)$$

For a fluid-side particle $z_e < 0$ and $W' < 0$, so $\lambda_\Delta > 0$; $\lambda_\Delta = 0$ when the support lies inside S or is disjoint from it.

Proof. $\nabla_{\mathbf{x}}^2 W(|\mathbf{x} - \mathbf{x}'|) = \nabla_{\mathbf{x}'}^2 W$, so Green's theorem gives $\oint_{\partial S} \partial_n W ds$ with $\nabla_{\mathbf{x}'} W = W'(r)\mathbf{y}/r$ and $\mathbf{n}_e \cdot \mathbf{y} = z_e$. If $\mathbf{x} \in S$ excise a small disk: its boundary flux is $2\pi\epsilon W'(\epsilon) \rightarrow 0$ since $W'(r) = O(r)$ at the centre, so there is no angle term; the outer support circle contributes nothing because $W'(1) = 0$. If the support lies inside S the boundary is outside the support, so the sum is empty; if disjoint, the chords are empty. $\square$

Numerically, Eq. (21) agrees with a ray quadrature of $\int_S \nabla^2 W$ over a non-convex polygon to $\leq 8 \cdot 10^{-11}$ for all four kernels, for a particle inside, outside and near an edge (`paper_edge_identities_k6.py`). [N]

Evaluation with the existing operators. Let $L = W'/r$. Then $W' = rL$, $W'' = L + rL'$ and $\nabla^2 W = W'' + W'/r = 2L + rL'$ in 2D [M]. With the density $\lambda[L] = \int_S L$ and the covariance $\text{Cov}[L] = \int_S \mathbf{y} \otimes \nabla_{\mathbf{x}} L$ (where $\nabla_{\mathbf{x}} L = -L'\mathbf{y}/r$ for $\mathbf{y} = \mathbf{x}' - \mathbf{x}$, hence $\text{tr Cov}[L] = -\int_S rL'$),

$$\lambda_\Delta = 2\lambda[L] - \text{tr Cov}[L]. \quad (22)$$

Registering L rather than $\nabla^2 W$ as the kernel is essential: inside a body the indicator contribution is $2 \cdot 1 - \text{tr}(\mathbb{K}) = 0$ and cancels (over all of $\mathbb{R}^2$, $\text{Cov} = \mathbb{K} \int L$ by parts, so $2 \int L - 2 \int L = 0$, as $\int \nabla^2 W = 0$ demands), whereas a directly registered $\nabla^2 W$ would give a spurious constant: the scene assumes $\int K = 1$ for a registered kernel, but $\int \nabla^2 W = 0$.

Wendland kernels: L is a multiple of an ordinary kernel. With $W = C_2 h^{-2} \widehat{W}(q)$ one finds $W'/r = C_2 h^{-4} \widehat{W}'(q)/q$ and

$$\widehat{W}'_{w2}/q = -20(1-q)^3, \quad \widehat{W}'_{w4}/q = -\frac{56}{3}(1+5q)(1-q)^5 = -\frac{56}{3}(6(1-q)^5 - 5(1-q)^6) \quad [\text{M}].$$

Normalise the shapes $(1-q)^3$ and $6(1-q)^5 - 5(1-q)^6$ as 2D kernels $K_\ell = C_\ell \text{shape}(q)/h^2$ with $2\pi C_\ell \int_0^1 q \text{shape} dq = 1$: since $\int_0^1 q(1-q)^3 dq = \frac{1}{20}$ and $\int_0^1 q(1+5q)(1-q)^5 dq = \frac{3}{56}$, $C_\ell = \frac{10}{\pi}$ (w2) and $\frac{28}{3\pi}$ (w4). Then $L = f K_\ell$ with

$$f = \frac{P C_2}{C_\ell h^2} = -\frac{14}{h^2} \quad (\text{w2: } P = -20, C_2 = \frac{7}{\pi}), \quad f = -\frac{18}{h^2} \quad (\text{w4: } P = -\frac{56}{3}, C_2 = \frac{9}{\pi}),$$

where P is the constant in front of the shape in the display above. [M]

9.2 The viscosity coefficient

Lemma 9.2. *For every normalised radial kernel in n dimensions, $\int_{\mathbb{R}^n} r W'(r) d\mathbf{x} = -n$; for $n = 2$, -2 .*

Proof. In 2D, $2\pi \int_0^1 r^2 W' dr = 2\pi ([r^2 W]_0^1 - 2 \int_0^1 r W dr) = -2$, since $2\pi \int r W dr = 1$ is the normalisation. (Likewise in n dimensions by one integration by parts.) [M] $\square$

The solver's artificial viscosity is, for fluid pairs, the pairwise sum (with $\text{fac} = \alpha c_0 h / \xi$ the combination of the scheme's constants)

$$\mathbf{a}_i = \text{fac} \sum_j \frac{m_j}{\bar{\rho}_{ij}} \frac{(\mathbf{v}_i - \mathbf{v}_j) \cdot (\mathbf{x}_i - \mathbf{x}_j)}{|\mathbf{x}_i - \mathbf{x}_j|^2} \nabla_i W_{ij}, \quad (23)$$

which we take from the code (`deltasph2d.py`) as given. Its continuum limit fixes the coefficient ν_{eff} :

Proposition 9.3 (Pairwise bulk term). *For a smooth velocity field the sum Eq. (23) converges to $\mathbf{a} = \frac{\text{fac}}{2(n+2)} (\nabla^2 \mathbf{v} + 2\nabla \nabla \cdot \mathbf{v})$; in 2D $\nu_{\text{eff}} = \text{fac}/8 = \alpha c_0 h / (8\xi)$ is the coefficient of $\nabla^2 \mathbf{v}$. [M (moments), N (lattice), A (form Eq. (23))]*

Proof. Replace $\sum_j m_j / \bar{\rho}_{ij}$ by $\int d\mathbf{x}'$, put $\mathbf{y} = \mathbf{x}' - \mathbf{x}$, $\delta \mathbf{v} = \mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})$. Then $\mathbf{v}_i - \mathbf{v}_j = -\delta \mathbf{v}$, $\mathbf{x}_i - \mathbf{x}_j = -\mathbf{y}$ and $\nabla_i W_{ij} = -W' \mathbf{y} / r$, so $a_i = -\text{fac} \int \frac{W'}{r^3} y_i y_k \delta v_k d\mathbf{x}'$. Taylor-expand $\delta v_k = y_l \partial_l v_k + \frac{1}{2} y_l y_m \partial_l \partial_m v_k + \dots$. Odd moments vanish by symmetry. For the quadratic term the isotropic fourth moment

$$\int y_i y_k y_l y_m F(r) d\mathbf{x} = \frac{\delta_{ik} \delta_{lm} + \delta_{il} \delta_{km} + \delta_{im} \delta_{kl}}{n(n+2)} \int r^4 F d\mathbf{x}$$

(check by contracting $i = k, l = m$) with $F = W'/r^3$ gives $\int r^4 F = \int r W' = -n$ by Lemma 9.2. Hence $a_i = -\text{fac} \cdot \frac{1}{2} \cdot \frac{-n}{n(n+2)} (\partial_l \partial_l v_i + 2\partial_i \partial_k v_k)$, which is the claim. $\square$

Numerically, for generic quadratic velocity fields and all four kernels the continuum integral reproduces $\frac{1}{8}(\nabla^2 \mathbf{v} + 2\nabla \nabla \cdot \mathbf{v})$ to 10^{-13} (cubic $2 \cdot 10^{-9}$, quadrature of the knot) and the lattice sum Eq. (23) with $h/\Delta x = 16$ to $\leq 1.2 \cdot 10^{-5}$ (`scripts/derivation_checks/paper_viscosity_limits.py`). The only moment of the kernel that survives is $\int r W'$, which is fixed by the normalisation: the coefficient does not depend on the kernel family. Note that the bulk term contains the extra part $2\nabla \nabla \cdot \mathbf{v}$; the wall terms below carry only the ∇^2 part, so wall and bulk share the Laplacian coefficient but are not one operator.

9.3 Free-slip wall term

A Laplacian estimate of the smoothed field, exact for the convolved field, is $\nabla^2 \mathbf{v}(\mathbf{x}) \simeq \int (\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})) \nabla^2 W(|\mathbf{x} - \mathbf{x}'|) d\mathbf{x}'$ (because $\int \nabla^2 W = 0$). (Equivalently, by $\int r W' = -n$, $\nabla^2 \mathbf{v} = -2 \int (\mathbf{v}' - \mathbf{v}) W'/r$, the Morris form; both estimators converge as $O(h^2)$, error ratios 4.0 per halving of h for w2, w4 and the cubic.) The solid part of the integral needs ghost values. With the free-slip mirror, constant normal $\mathbf{n}$ per particle (pointing *into* the wall) and $u_n = (\mathbf{v} - \mathbf{v}_w) \cdot \mathbf{n}$, the ghost velocity is the mirror image, $\mathbf{v}_{\text{ghost}} - \mathbf{v} = -2u_n \mathbf{n}$, constant over the solid. Then

$$\mathbf{a}_{\text{wall}} = \nu_{\text{eff}} \frac{m_{\text{wall}}}{\rho_i} \int_S (\mathbf{v}_{\text{ghost}} - \mathbf{v}) \nabla^2 W d\mathbf{x}' = -2\nu_{\text{eff}} \frac{m_{\text{wall}}}{\rho_i} u_n \lambda_\Delta \mathbf{n}. \quad (24)$$

Only the normal relative velocity is damped, and it is damped when moving into the wall ($u_n > 0$, $\lambda_\Delta > 0$). This differs from the pairwise free-slip form near the wall: $\lambda_\Delta \rightarrow 0$ linearly as the particle reaches the wall (the support loses half its disk and the odd mirror field vanishes), while the pairwise coefficient does not, by design. [A]

9.4 No-slip flux term

Following Chiron et al. [4], $\int_{\text{chord}} W \, ds$ over the cut face is the per-edge channel of $\nabla \lambda$ (Theorem 3.4). Define $|G| = m_{\text{wall}} \int_{\text{chord}} W \, ds$ and

$$\mathbf{a}_{\text{wall}} = -2\nu_{\text{eff}} (\mathbf{v}_i - \mathbf{v}_w) |G| / (\rho_i d_b), \quad d_b = \max(d, d_{\min}). \quad (25)$$

This is a *modelling* step: it replaces the normal derivative at the wall by the one-sided difference $(\mathbf{v} - \mathbf{v}_w)/d$ (neglecting the tangential gradient, Chiron et al. Eq. 89–90) and uses the viscous coefficient ν_{eff} of the scheme (an artificial coefficient, about 3000 times the kinematic viscosity of water in the tested tank). The factor 2 is the same 2 as in the Morris form above ($\nabla^2 \mathbf{v} = 2 \int (\mathbf{v} - \mathbf{v}') W' / r$, a consequence of $\int r W' = -n$), which in Chiron’s operator multiplies the fluid sum and the wall surface term alike: their Eq. (92) is $\frac{2}{\gamma_i} \left[\sum_j \omega_j (u_i - u_j) \frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|^2} \cdot \nabla_i W_{ij} + \sum_s s_s \frac{u_i - u_s}{d_n} W_{is} \right]$, and in their derivation (Eq. (84)–(91)) the 2 in front of the surface term comes from $\nabla f(\mathbf{x}) \cdot \mathbf{n} \approx \nabla f(\mathbf{y}) \cdot \mathbf{n}$ in the boundary flux $\int_{\partial\Omega} (\nabla f(\mathbf{x}) \cdot \mathbf{n} + \nabla f(\mathbf{y}) \cdot \mathbf{n}) W \, dS$. (With $\mathbf{n}$ outward from the fluid their d_n is signed so that the wall term opposes $u_i - u_s$, as in the formula above.) None of this follows from the preceding calculus, so only the identity $|G| = m_{\text{wall}} \int W \, ds$ is a theorem here. [A]

10 Verification ledger

Table 2 lists each result with the strongest evidence behind it.

Reproduce (all Maple lines must read PASS):

- `make -C paper` builds the paper;
- `maple -q maple/20_paper_audit.mpl`: symbolic and 40-digit checks of this paper (110 PASS lines);
- `bash maple/run_edge.sh`: the project’s edge-calculus checks (61 lines at the time of writing);
- numerical checks in `scripts/derivation_checks/` (compare with the quoted numbers):
 - `paper_ngon_orders.py`, `paper_curvature_table.py` (seconds);
 - `paper_curvature_variable.py`, `paper_viscosity_limits.py` (seconds);
 - `paper_edge_identities_k6.py` (about 4 minutes).

A Audit

A.1 The existing derivation notes

The mathematical content of `docs/derivation.md` and `docs/derivations/*.md` was re-derived and re-checked (Maple, exact arithmetic or 40 digits). **No error was found in any final formula.** The following are presentation or scope issues, in decreasing importance.

1. **Sign convention in `laplacian-wall.md` §3.** Step 1 defines $\mathbf{y} = \mathbf{x} - \mathbf{x}'$, Step 2 then writes $\text{tr Cov} = \int r L'$ but the header and the conclusion need $\text{tr Cov} = -\int r L'$ (true for $\mathbf{y} = \mathbf{x}' - \mathbf{x}$). The final identity Eq. (22) is correct with $\mathbf{y} = \mathbf{x}' - \mathbf{x}$; the intermediate line is not.
2. **Tier-3 gradient scope (`tier3-curvature-2d.md` §1):** “ $\nabla_{\mathbf{x}} \lambda = \partial_d \lambda \hat{\mathbf{n}}$ ” holds only for constant curvature; for general curves a tangential term $\kappa' F_1$ appears at second order (Remark 6.2). The value expansion needs no κ' term.

3. **J_m structure** (`derivation.md` §3.2): odd m are $\text{poly}(d^2)\sqrt{1-d^2}$ (no constant), by Eq. (9).
4. **Edge value identity**, `tet-face-edge-chain.md`: status [P] is accurate; the divergence step is nonetheless a theorem (Theorem 8.1) and could be tagged as such.
5. **ν_{eff} and the no-slip factor 2** (`laplacian-wall.md` §3.5, `noslip-wall.md` §3.3) are tagged [V] in the notes because the *implementation* matches the stated formula; as mathematics, the first depends on the pairwise term (not re-derived) and the second is a modelling choice.
6. **Block identity wording**. `truncated-monomials.md` correctly notes that block-by-block gradients need no cancellation between blocks; this is Theorem 3.4. The known bug in `scripts/derivation_checks/monomial_edge_forms.py` (cubic correction -4σ instead of -8σ) remains in the unused script.

Not independently audited here: the solver-side statements in `docs/deltasph-*.md` and `docs/dfsph-validation.md`, which are validation reports rather than derivations.

A.2 The first draft of this paper

Each derivation of the first draft was re-done independently by hand, the symbolic script re-run, and the numerical claims recomputed (the curvature table and the cubic value at $d = \frac{1}{2}$ with quadrature, the polygon orders with a ray integration against the disk shell formula). Again no final formula was wrong, but the draft contained the following errors or gaps, now corrected in the text, in decreasing importance:

1. **Remainder in Theorem 6.1.** The draft said that φ is C^2 with φ'' Lipschitz for all four kernels and argued with a jump set of measure $O(\kappa)$. φ'' is in fact singular at $\sigma = 0$ ($15/r$ for w2, $\frac{9}{4}/r$ for the cubic). The result is unaffected because $\sigma \geq d_0^2 - O(\kappa) > 0$, and where φ'' is Lipschitz the remainder bound is immediate, so the measure argument was unnecessary. The profile list “powers of r down to r^{-1} for Wendland” was likewise true only for w2.
2. **Block weights.** The draft called $\frac{8}{7}$ and $-\frac{1}{7}$ both the indicator weight and the arctan coefficient; they are the indicator weights $2\pi M_j(r_j)$, and the arctan coefficient is $M_j(r_j) = w_j/2\pi$.
3. **Edge crossing (Section 3.5).** The draft wrote that the arctan term jumps by $+1$ “in the sense that cancels” the indicator jump; in Eq. (12) the chord term jumps by -1 (the arctan appears with a minus sign in Eq. (14)), which is what cancels. The vertex convention is now justified by an explicit limit.
4. **Primitives for odd j .** “The same with J ” hid that the binomial runs over $p = \lfloor j/2 \rfloor$ and the exponent is $m + 2i + 2$; written out in Proposition 3.1(c).
5. **Cubic value at $d = \frac{1}{2}$.** Closed form correct, decimal wrong: 0.03744, not 0.03748.
6. **Proposition 3.9.** The draft called the half-plane check a proof of the edge calculus against the shell calculus; the proof tests the gradient identity, and the value identity enters only through the closing at infinity. Reworded.
7. **ν_{eff} .** Marked “not re-derived”; it is now derived from the solver’s pairwise form (Proposition 9.3). The form itself is still taken from the code. The factor 2 of the no-slip term is traced to the Morris normalisation instead of “by analogy”, with the caveat that Chiron’s Eq. 92 is quoted from the project notes.

8. **Scaling lemma.** $m_\alpha(h) = h^{|\alpha|}m_\alpha(1)$ “after scaling all lengths” was ambiguous; now $m_\alpha(d; h) = h^{|\alpha|}m_\alpha(d/h; 1)$.
9. **Variable-curvature gradient.** “First-order accurate” replaced by the actual statement: an $O(\kappa^2)$ tangential error unless $\kappa'F_1$ is added.
10. **Polygon orders (Proposition 5.1).** The “hence proportional to the mean deviation” step is now an argument (ring integral, two integrations by parts) with the exact hypothesis ($W \in C^2$).
11. **Variable-curvature gradient (Remark 6.2).** The leading tangential term $\kappa'F_1$ is right, but the draft stopped there; measured numerically, the next term is $\kappa\kappa'(2F_2 - dF_1)$, with the $-dF_1$ from $\partial_x\kappa = \kappa'/(1 + \kappa d)$.

Presentation only: the symbols R (cut radius versus body radius), λ_i (barycentric versus kernel integral), J_m (planar family versus edge primitive), Φ (compact potential versus kernel profile), $\Delta\lambda$ (read as a difference) and H versus h each had two meanings; see Section 1.4. λ_Δ , P , c and C_ℓ were also undefined where used.

A.3 Independent numerical verification of the 2D claims

The 2D claims that were tagged A or checked only inside the project were re-verified with code that does not use the project library (`scripts/derivation_checks/paper_*.py`): moment recursion for all $|\alpha| \leq 6$ and gradient, g_α and Green-form identities against ray quadrature (Sections 3 and 9); the pairwise-viscosity limit and the two Laplacian estimators (Section 9); the variable-curvature statements and the disk-series orders (Sections 6 and 7); the polygon orders over $N = 12, \dots, 96$ (Section 5). The Chiron et al. Eq. (92) was read in the original. Not covered: everything in Section 8 and the 3D strand, curved centre line and sphere statements; the pairwise form Eq. (23) itself, which is quoted from the solver; and the free-slip and no-slip wall terms as models.

References

- [1] H. Wendland, Piecewise polynomial, positive definite and compactly supported radial functions of minimal degree, *Adv. Comput. Math.* 4 (1995) 389–396.
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- [3] R. Winchenbach, A. Kolb, arXiv:2507.21686 (2025), Sec. 6.2 (FEM-weighted boundary integrals, validation problems).
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result	tag	evidence
Kernel constants, block weights	M	20_paper_audit.mpl
Planar 3D closed forms	M	same (cubic both branches, degrees 8, 11, 14)
Planar 2D by-parts, $\mathcal{J}_m$ recursion, w2 explicit form, cubic join	M	same, 40 digits where numeric
Sphere cap area, planar limit	M	same
Edge primitives	M	10_edge_primitives.mpl
Blocks, value/gradient, half plane	M,N	11,13, tests/edge/test_value.py, test_gradient.py
Moment recursion $k \leq 4$ (symbolic), $k \leq 6$ (numeric)	M,N	12_moments_recursion.mpl; paper_edge_identities_k6.py
FEM weights $p \leq 3$	N	tests/edge/test_fem*.py
Polygon-to-disk orders	A,N	argument in Proposition 5.1; test_tier2.py; paper_ngon_orders.py ($N = 12-96, \rightarrow 0.377 a^4$)
Curvature expansion	M,N	14_tier3_2d.mpl, 20_paper_audit.mpl; table reproduced by paper_curvature_table.py
Disk/strip series (2D)	M,N	20_paper_audit.mpl, test_tier4.py, paper_curvature_variable.py
Variable curvature (even in κ' ; tangential gradient)	N	paper_curvature_variable.py
Wall Laplacian Green form	M,N	identities in 20_paper_audit.mpl; tests/scene/test_viscosity_scene.py
ν_{eff} from the pairwise term	M, N, A	Proposition 9.3: moments in 20_paper_audit.mpl, lattice in paper_viscosity_limits.py; the pairwise form Eq. (23) is quoted from the code
Free-slip wall term	A	Eq. (24): ghost-value argument, model
No-slip coefficient	A	modelling choice
3D face identity	M (div.)	divergence step only
3D evaluation, curved centre line	P	open

Table 2: Verification ledger.